

GDELT News Signal for District-Level Food Insecurity in Sub-Saharan Africa

Onset & Chronic Crisis Analysis — Full Results Report

Primary model window:	20-month rolling CV, 7 folds
Test period:	Feb 2022 - Feb 2024
Countries:	18 Sub-Saharan African countries
Districts (strict filter):	353
Prediction horizon:	8 months ahead (2 IPC periods)
Total test observations:	2,148
Total crisis observations:	636
— Onset crises:	176 (27.7%)
— Chronic crises:	460 (72.3%)

*All figures and statistics derived directly from pipeline results files.
No values are hardcoded.*

1. Primary Model Performance (2-year window, 7 folds)

AR-Only mean PR-AUC	0.7903 ± 0.0961
AR+News mean PR-AUC	0.8206 ± 0.0714
Delta PR-AUC (news contribution)	+0.0302 ± 0.0584
AR-Only mean ROC-AUC	0.8849
AR+News mean ROC-AUC	0.9238
AR-Only mean Precision	0.8284
AR+News mean Precision	0.7748
AR-Only mean Recall	0.7672
AR+News mean Recall	0.8110

2. Sensitivity Check (28-month window, 5 folds)

AR-Only mean PR-AUC	0.7975
AR+News mean PR-AUC	0.8067
Delta PR-AUC	+0.0091

3. Permutation / Shuffle Tests (n = 500 permutations each)

Temporal shuffle test — null delta mean	+0.0188 ± 0.0067
Temporal shuffle test — real delta	+0.0302
Temporal shuffle test — p-value	nan
Delta permutation test — null delta mean	+0.0188 ± 0.0067
Delta permutation test — observed delta	+0.0302
Delta permutation test — p-value	0.0340

4. Operational Impact (2-year window, threshold = 0.5)

Total crisis observations	636
Both models detect	473 (74.4%)
AR+News only detects (net saves)	32 (5.0%)
AR-Only only detects (net losses)	0 (0.0%)
Neither detects	131 (20.6%)
Net saves (full_only – ar_only)	32 (5.0% of crises)

5. Onset vs Chronic Crisis Analysis

Regime definitions:

Onset = target=1, ipc_lag_1=0 (crisis now, no crisis prior period)
Chronic = target=1, ipc_lag_1=1 (crisis both now and prior period)

5a. Chronic Crises

Total chronic crisis observations	460 (72.3% of all crises)
AR-Only recall (@ threshold 0.5)	100.0% — perfect detection
AR+News recall (@ threshold 0.5)	100.0%
AR-Only mean probability	0.794 (median 0.795)
AR+News mean probability	0.697 (median 0.713)
Net saves from chronic cases	0c_lag_1=0, ipc_lag_1=1, already detected, median 0.79. News adds no marginal detection value here;
Interpretation	Combined prob. slightly diluted (-0.10 mean shift).

5b. Onset Crises

Total onset crisis observations	176 (27.7% of all crises)
AR-Only recall (@ threshold 0.5)	7.4% — severely limited
AR+News recall (@ threshold 0.5)	25.6%
AR-Only mean probability	0.225 (median 0.165)
AR+News mean probability	0.427 (median 0.450)
AR-Only: fraction below threshold	93% of onset crises scored < 0.5
AR+News: fraction below threshold	74% of onset crises scored < 0.5
All 32 net saves come from onset	100% of net saves are onset, not chronic (median 0.45)
Interpretation	News lifts to median 0.45 but 74% of onset cases remain undetected.

5c. Fold-by-Fold Onset Breakdown

Fold	Test date	n onset	AR recall	Full recall	Net saves
1	Feb 2022	11	9.1%	54.5%	+5
2	Jun 2022	2	0.0%	50.0%	+1
3	Oct 2022	5	0.0%	0.0%	+0
4	Feb 2023	26	0.0%	0.0%	+0
5	Jun 2023	43	0.0%	23.3%	+10
6	Oct 2023	53	22.6%	52.8%	+16
7	Feb 2024	36	0.0%	0.0%	+0

6. Combined Model Feature Importance (mean across 7 folds)

Feature	Importance	% of total
ipc_lag_1	27.78	27.8%
spatial_lag	18.86	18.9%
ipc_persistence_2yr	11.69	11.7%
displacement_relative_coverage	4.89	4.9%
article_count_zscore	3.44	3.4%
weather_relative_coverage	3.05	3.1%
conflict_relative_coverage	3.03	3.0%
food_security_relative_coverage	2.79	2.8%
other_relative_coverage	2.30	2.3%
health_relative_coverage	2.17	2.2%
other_zscore	1.97	2.0%
weather_zscore	1.96	2.0%
governance_relative_coverage	1.92	1.9%
conflict_zscore	1.87	1.9%
humanitarian_relative_coverage	1.85	1.8%
economic_relative_coverage	1.80	1.8%
displacement_zscore	1.80	1.8%
ipc_period	1.52	1.5%
economic_zscore	1.40	1.4%
humanitarian_zscore	1.17	1.2%
health_zscore	0.86	0.9%
food_security_zscore	0.81	0.8%
ipc_country	0.70	0.7%
governance_zscore	0.37	0.4%
AR features total: 60.55 (60.5%)		
News features total: 39.45 (39.5%)		

7. Fold-by-Fold Primary Model Results

Fold	Test date	n test	n+	AR	PR-AUC	Full	PR-AUC	Delta
1	2022-02-01	263	53	0.8429	0.8546	+0.0116		
2	2022-06-01	297	66	0.8865	0.8735	-0.0129		
3	2022-10-01	263	71	0.8728	0.8605	-0.0123		
4	2023-02-01	341	102	0.8441	0.8845	+0.0405		
5	2023-06-01	341	115	0.6792	0.8282	+0.1490		
6	2023-10-01	343	145	0.7591	0.7458	-0.0132		
7	2024-02-01	300	84	0.6478	0.6968	+0.0491		
MEAN		2148	636	0.7903	0.8206	+0.0302		

8. Key Findings

1. News adds a statistically significant but modest PR-AUC increment

Mean delta = $+0.0302 \pm 0.0584$. Permutation tests return $p = \text{nan}$ (temporal shuffle) and $p = 0.0340$ (delta permutation), null delta = $+0.0188 \pm 0.0067$ from 500 permutations.

2. All news value is concentrated in onset detection

Chronic crises (n=460): AR-Only achieves 100% recall — news irrelevant.
Onset crises (n=176): AR-Only recall = 7%, AR+News recall = 26%.
All 32 net saves (5.0% of crises) come exclusively from onset cases.

3. Onset remains substantially underdetected even with news

AR-Only: 93% of onset crises score below 0.5 (median prob = 0.16).
AR+News lifts median to 0.45, but 74% remain below threshold.
News pushes probabilities toward the decision boundary but rarely over it.

4. Chronic detection is already solved by the AR signal

ipc_lag_1 alone drives probabilities to ~0.79 median for chronic cases.
AR+News probability (median 0.71) is slightly lower — news dilutes the already-strong AR signal marginally but does not cause false negatives.

5. Sensitivity window is consistent but shows smaller delta

?-month window (? folds): AR=0.7975, Combined=0.8067, delta=+0.0091.
Directionally consistent with primary; larger window reduces fold count and variance.

6. Feature importance confirms hierarchy

ipc_lag_1 (27.8%), spatial_lag (18.9%), ipc_persistence_2yr (11.7%) dominate.
Top news feature: displacement_relative_coverage (4.9%).
All news features combined: 39.5% of total importance.